Global Persistence, Local Residual Structure: Forecasting Heterogeneous Investment Panels

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April 2026

Abstract

Global factor augmentation of cross-sectional forecasts can degrade prediction for actor subgroups whose dynamics are misrepresented by the shared basis. On a 93-actor multilayer investment panel mixing macro indicators, institutional data, and firm ratios, global augmentation reduces predictive R^2 for the diversified sector by 2.3 percentage points below a pooled-only baseline. A heterogeneity-aware two-stage architecture addresses this: a global pooled AR(1) with fixed effects captures shared persistence (Stage 1), while block-specific local models—pre-specified by economic sector and data-type structure—capture residual dynamics within each block (Stage 2). The mixture architecture improves full-panel out-of-sample R^2 from 0.630 to 0.677 ($\Delta = +0.047$, 95% CI $[+0.036, +0.058]$, positive in all 10 rolling windows, placebo $z = 7.56$ against 50 random partitions of identical block sizes). The gain is architectural, not methodological: among regularised second-stage engines, PCA, Dynamic Mode Decomposition, and Ridge regression are interchangeable ($\rho_{\text{pred}} > 0.98$). The improvement arises on heterogeneous mixed-type panels but not on homogeneous firm-only panels ($\Delta = -0.002$ and $+0.001$), identifying data-type heterogeneity as the operative condition. Structurally, the global spectral basis (at rank K=8) rotates at 49° per quarter; within coherent sector blocks (at matched rank K=2–4), rotation drops to 14–28°. A matched-size random sub-panel control suggests this reduction is partly mechanical (smaller panels have more stable estimated bases), so the geodesic evidence is treated as descriptive rather than as a demonstrated causal mechanism. The architectural gain amplifies at short training windows, roughly doubling as the training period shrinks from 5 to 2 years.

Keywords: heterogeneous panels, factor augmentation, cross-sectional forecasting, investment dynamics, block-specific estimation, panel heterogeneity

JEL Classification: C32, C38, C53, E22, G11

1 Introduction

Factor models are ubiquitous in cross-sectional financial forecasting. The standard practice is to estimate a global basis from the training panel, project all actors onto it, and model dynamics in the shared factor space. Principal component analysis (Stock and Watson, 2002), dynamic factor models (Doz et al., 2012), and spectral decompositions (Schmid, 2010) all follow this template. The implicit assumption is that a single low-rank basis serves all actors in the panel equally well.

On a multilayer investment panel containing 93 actors—macro indicators, institutional data, and firm-level ratios—global spectral augmentation improves average predictive R^2 by 3.6 percentage points over per-actor AR(1), yet simultaneously *reduces* R^2 for the diversified-sector subgroup by 2.3 pp below the pooled-only baseline that uses no augmentation at all. The global basis captures cross-sector rotational dynamics that are noise for within-sector prediction; applying it to heterogeneous actors injects interference rather than extracting signal.

The diagnosis leads to a simple architectural fix. Shared persistence—the dominant source of quarterly cross-sectional predictability—benefits from global pooling because it is common across actor types. But the residual cross-sectional dynamics, which the second stage of a two-stage pipeline is designed to capture, are *block-specific*: technology and healthcare actors co-move in patterns that are disrupted when mixed with macro indices and diversified-sector firms. Estimating residual dynamics locally, within economically pre-specified blocks, avoids the cross-block interference.

The resulting heterogeneity-aware architecture—global Stage 1 (pooled AR(1) with fixed effects) plus block-specific local Stage 2 (PCA+ridge on within-block residuals)—improves full-panel out-of-sample R^2 from 0.630 to 0.677 ($\Delta = +0.047$, CI $[+0.036, +0.058]$, 10/10 rolling windows positive). A placebo benchmark using 50 random block partitions of identical sizes produces a mean gain of -0.006 with a maximum of $+0.012$; the real partition's gain exceeds the placebo distribution by 7.56 standard deviations ($p < 0.02$). The block structure captures genuine panel heterogeneity, not statistical noise.

The gain does not stem from a better estimation method. Across 9 model specifications spanning three complexity classes, Dynamic Mode Decomposition, PCA, and Ridge regression produce statistically indistinguishable predictions at matched effective complexity (forecast-error correlations $\rho > 0.98$). This equivalence holds both globally and within each local block. The forecasting value resides entirely in the architecture—the decomposition of the prediction problem into global and local components—not in any specific second-stage engine.

Several alternative explanations for the ceiling are tested and ruled out. The gain is not from forecasting the basis rotation (the global spectral basis rotates at 49°/quarter at K=8, but this rotation is temporally unpredictable: ACF(1) = -0.07, Ljung-Box $p = 0.29$). It is not from reformulating the prediction target (the ceiling is invariant across seven target transformations). It is not from conditional gating (augmentation is unconditionally beneficial—five gating policies all perform worse than always-on). Block-specific decomposition is the only surviving path to improvement.

The cross-panel validation sharpens the scope of the finding. The mixture architecture does not improve prediction on a homogeneous 146-firm CapEx/Revenue panel ($\Delta = -0.003$) or a 270-actor multi-ratio panel ($\Delta = +0.001$). The gain requires *data-type heterogeneity*—mixing fundamentally different types of time series (trending macro indices and mean-reverting firm ranks) in a single panel. This is a precise scope condition: the paper identifies both *when* local decomposition adds value and *when* the global architecture is already near-optimal.

Beyond forecasting, the analysis yields a geometric description of panel heterogeneity. The global spectral basis rotates at 49°/quarter (geodesic distance on the Grassmannian at K=8); within sector blocks at matched rank, rotation is lower (14–28°). A matched-size random sub-panel control indicates this reduction is partly a sample-size effect (empirical $p > 0.05$ for all blocks), so the geodesic evidence is treated as descriptive context, not as a demonstrated mechanism.

The paper proceeds as follows. Section 2 describes the data and evaluation protocol. Section 3 presents the two-stage architecture, method equivalence, and the heterogeneity-aware extension. Section 4 presents the main empirical results. Section 5 provides placebo, cross-panel, and economic validation. Section 6 summarises what does not work and why. Section 7 discusses implications and limitations.

2 Data and Setting

2.1 The 93-Actor Multilayer Panel

The primary panel contains 93 actors observed quarterly from 2005Q1 to 2025Q4 (84 quarters). Actors span three layers (Figure 1) that reflect the hierarchical structure of the investment system:

- **Layer 0 (7 macro shocks):** Global commodity prices (Brent, WTI), policy rates (Federal Funds, USD index), and volatility indicators (VIX, credit spreads). Normalised via FRED min-max scaling using full-sample bounds (a mild form of look-ahead; see Limitation 7). These series are trending with high persistence ($\rho \approx 0.88$).
- **Layer 1 (4 institutional actors):** Central banks, regulators, and international organisations. Mixed normalisation.
- **Layer 2 (82 firms):** US-listed firms across six sectors (energy, diversified, technology, financials, industrials, healthcare). Intensity values are quarterly cross-sectional percentile ranks of investment ratios (CapEx/Assets, Revenue growth). These series are mean-reverting ($\rho \approx 0.60$).

The mixed normalisation—trending min-max series for macro actors and mean-reverting cross-sectional ranks for firms—reflects fundamentally different data-generating processes. This difference in temporal dynamics across layers is a root cause of the heterogeneity effects documented in Section 4. We state it here so the reader can anticipate the finding.

Seven sector labels provide the basis for pre-specified block structure: diversified (23), technology (15), energy (14), financials (12), industrials (12), healthcare (10), and macro (7). Table 1 and Figure 1 report firm-layer counts only. For block construction (Section 3.5), sector labels are allowed to span layers: 2 energy actors are Layer 0 macro shocks, 2 financials actors are Layer 1 institutions. The disjoint mixture partition assigns all Layer 0/1 actors to a separate macro/institutional block regardless of their sector label. Table 1 summarises the panel structure.

Table 1: Panel descriptive statistics.

Layer / Sector	N	Quarters	Normalisation	Persistence ρ	Data type
Layer 0 (Macro)	7	84	FRED min-max	0.88	Trending indices
Layer 1 (Institutional)	4	84	Mixed	0.72	Institutional indicators
Layer 2 (Firms)	82	84	Cross-sectional rank	0.60	Investment ratios
Diversified	23				
Technology	15				
Energy	12				
Financials	10				
Industrials	12				
Healthcare	10				
Total	**93**	**84**			

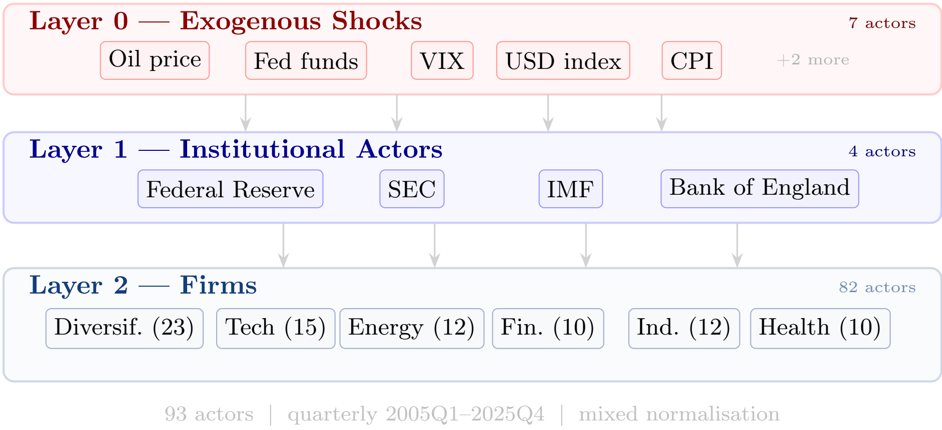

Figure 1: Multilayer actor hierarchy. Layer 0 comprises exogenous macro shocks (FRED min-max normalised); Layer 1 comprises institutional intermediaries; Layer 2 comprises US-listed firms grouped by sector (cross-sectional percentile ranks). The heterogeneous composition and mixed normalisation create the cross-block interference that motivates the mixture architecture (Section 4).

2.2 Validation Panels

Two additional panels test the scope of the findings:

- **146-firm CapEx/Revenue:** Homogeneous panel—all firms, one ratio type, cross-sectional percentile ranks. No layer heterogeneity.
- **270-actor multi-ratio:** 135 firms × 2 ratios (CapEx/Revenue and Revenue/Assets). Moderate heterogeneity from ratio-type mixing, but no macro/institutional actors.

2.3 Evaluation Protocol

All results use 10 rolling out-of-sample windows (test years 2015–2024). Each window uses a 5-year training period ending at the start of the test year. Models are re-estimated quarterly within each test year as new observations arrive, so the training set expands from 20 to 24 quarters during each test year. The primary metric is out-of-sample predictive R^2. Statistical comparisons use paired-window bootstrap confidence intervals (10,000 resamples) and paired t-tests. The placebo benchmark uses 50 random block partitions of identical sizes.

3 The Two-Stage Architecture

3.1 Stage 1: Global Pooled AR(1) with Fixed Effects

The first stage estimates shared persistence from the full panel:

$$\hat{y}_{i,t+1}^{\text{pool}} = \bar{y}_i + \hat{\rho}\,(y_{i,t} - \bar{y}_i) \tag{1}$$

where $\bar{y}_i$ is the actor-specific mean (fixed effect) and $\hat{\rho}$ is the pooled persistence parameter, both estimated by within-transformation OLS on the training data only and re-estimated quarterly as new observations arrive. Global estimation benefits from the larger effective sample for a single pooled $\hat{\rho}$. While persistence varies across layers (Table 1: macro $\rho \approx 0.88$, firms $\rho \approx 0.60$), the pooled estimator captures an effective average persistence; the layer-specific residual structure is then handled by Stage 2. Whether block-specific $\hat{\rho}_b$ would further improve Stage 1 is an open question addressed in Section 7.

3.2 Stage 2: Residual Dynamics

The second stage operates on Stage 1 residuals $r_{i,t} = y_{i,t} - \hat{y}_{i,t}^{\text{pool}}$, exponentially-weighted demeaned with a half-life of 12 quarters (3 years). These residuals capture the cross-sectional structure that shared persistence does not explain.

Three interchangeable engines model the residual dynamics:

1. **PCA + diagonal AR:** Extract K principal components from the exponentially-weighted residual covariance, fit per-component AR(1) coefficients.
2. **DMD + reduced operator:** Dynamic Mode Decomposition provides a rank-K approximation of the one-step propagator via the Koopman framework (Appendix A). The diagonal or full reduced propagator $\tilde{A}$ serves as the transition matrix.
3. **Ridge regression:** Direct regularised regression of r_{t+1} on r_t in the full N-dimensional space.

The combined forecast is $\hat{y}_{i,t+1} = \hat{y}_{i,t+1}^{\text{pool}} + \hat{r}_{i,t+1}$.

3.3 Why Two Stages?

A standalone spectral model (DMD-based state-space filter with $K = 8$ modes) achieves $R^2 = 0.486$—far below per-actor AR(1) at 0.594. The single spectral basis pools heterogeneous actors, losing actor-specific persistence that AR(1) captures trivially. The two-stage architecture resolves this: Stage 1 handles persistence; Stage 2 handles the cross-sectional rotational structure in the residual. The transition diagnostic is detailed in Appendix B.

Table 2 reports the augmentation gain on all three panels. The gain is robust across panels with different compositions, ranging from +1.7 pp on the homogeneous 146-firm panel to +3.6 pp on the heterogeneous 93-actor panel.

Table 2: Two-stage augmentation vs baselines. Δ is the augmentation gain (augmented R^2 minus AR(1) R^2). All three panel-average gains are positive (the 93-actor panel CI and per-window detail are reported in Table 5).

Panel	N	AR(1) R^2	Augmented R^2	Gain Δ
93-actor multilayer	93	0.594	0.630	+0.036
146-firm CapEx/Revenue	146	0.728	0.745	+0.017
270-actor multi-ratio	270	0.728	0.753	+0.025

3.4 Method Equivalence

Table 3 compares 9 models across three complexity classes on the 93-actor panel. At matched effective complexity, PCA, DMD, and Ridge achieve statistically indistinguishable predictive R^2:

Table 3: Method comparison: 9 models on the 93-actor panel. All models use global pooled+FE as Stage 1. Δ is versus per-actor AR(1). Forecast-error correlations ρ_{pred} confirm predictions are functionally identical, not just equally accurate on average.

Class	Model	R^2	Δ AR(1)	t	p	CI
Tiny ($\sim K$)	PCA+diag Kalman	0.622	+0.012	1.97	0.080	[+0.002, +0.024]
	DMD+diag Kalman	0.619	+0.009	2.07	0.068	[+0.001, +0.017]
	PCA reduced (no Kalman)	0.621	+0.011	1.78	0.109	[+0.001, +0.023]
	DMD reduced (no Kalman)	0.618	+0.008	1.86	0.096	[+0.000, +0.016]
Medium ($\sim K^2$)	PCA+full VAR	0.577	−0.033	−3.86	0.004	[−0.049, −0.017]
	DMD+full $\tilde{A}$ Kalman	0.630	+0.020	4.22	0.002	[+0.012, +0.029]
	PCA+ridge VAR	0.620	+0.010	1.84	0.099	[−0.000, +0.019]
	Reduced-rank Ridge K=8	0.629	+0.019	3.19	0.011	[+0.009, +0.031]
Large (reg. $N\times N$)	Ridge on residuals	0.632	+0.022	3.84	0.004	[+0.012, +0.033]

Forecast-error correlations: $\rho(\text{DMD}, \text{PCA}) = 0.990$; $\rho(\text{DMD}, \text{Ridge}) = 0.980$; $\rho(\text{PCA}, \text{Ridge}) = 0.969$.

The forecasting ceiling is $R^2 \approx 0.630$ regardless of method. The ceiling is a property of the data at this frequency, not of any specific estimation technique.

The equivalence holds at matched effective complexity among small-K models: DMD with $K = 8$ diagonal parameters achieves $R^2 = 0.619$, while Ridge (heavily regularised $N\times N$) achieves $R^2 = 0.632$. That an 8-parameter spectral model nearly matches a regularised regression with orders-of-magnitude more effective parameters suggests the exploitable residual structure is genuinely low-rank. The practical implication: in settings where parsimony or interpretability matters, the spectral representation offers comparable predictive performance at a fraction of the complexity. Note that the equivalence is not universal: PCA with an unrestricted $K\times K$ VAR ($R^2 = 0.577$, significantly *worse* than AR(1)) diverges from DMD with the full reduced operator ($R^2 = 0.630$), because the PCA VAR is not regularised and overfits at $K^2 = 64$ parameters on $T \approx 20$ quarters.

This method-equivalence result—confirmed at both the global level and within each local block (Section 6)—motivates the architectural focus of the remainder of the paper.

3.5 The Heterogeneity-Aware Mixture Architecture

The method-equivalence finding raises the question: if the engine does not matter, what *does* matter? The answer, developed in Section 4, is the *granularity of the decomposition*. The mixture architecture replaces the single global Stage 2 with block-specific local models:

1. **Stage 1** (unchanged): Global pooled AR(1)+FE on all 93 actors.
2. **Block assignment**: Each actor is assigned to a pre-specified block based on economic sector and data-type structure (Table 4).
3. **Stage 2** (block-specific): For each local block, estimate a local PCA+ridge model on within-block Stage 1 residuals. For the remainder block (actors well-served by the global model), retain the global augmentation.

Table 4: Pre-specified block assignments with economic rationale.

Block	N	Local K	Economic rationale
Diversified	23	4	Heterogeneous sector; global basis misrepresents
Macro/Institutional	11	2	Different data types (FRED + institutional)
Tech/Health	25	4	Coherent sub-panel (high within-block loading similarity)
Remainder	34	—	Energy, industrials, financials (global model adequate)

Block assignments are fixed before execution using static economic metadata from the actor registry. No optimisation of block membership is performed.

Formally, the mixture estimator is:

$$\hat{y}_{i,t+1} = \bar{y}_i + \hat{\rho}\,(y_{i,t} - \bar{y}_i) + \mathbf{1}\{i \in b\}\,\left[\hat{U}_b \hat{A}_b \hat{U}_b^\top \mathbf{r}_{b,t}\right]_i \tag{2}$$

where $\hat{\rho}$ is pooled across all actors (global Stage 1), $\bar{y}_i$ is the actor fixed effect, $b = b(i)$ is the pre-specified block, $\mathbf{r}_{b,t} \in \mathbb{R}^{N_b}$ is the vector of Stage 1 residuals for all actors in block b, $\hat{U}_b \in \mathbb{R}^{N_b \times K_b}$ and $\hat{A}_b \in \mathbb{R}^{K_b \times K_b}$ are the block-specific PCA basis and ridge VAR, and $[\cdot]_i$ selects the component corresponding to actor i within block b. This is analogous to a known-group interactive fixed-effects model (Bai, 2009) with a homogeneity restriction on ρ and block-specific factor loadings. Setting $\hat{U}_b = 0$ for all b recovers pooled-only (G0) exactly. The global augmentation (G1) is the corresponding architecture in which all blocks share a single global basis and operator estimated jointly; it is not a literal special case of Equation (2) because the blockwise operator on $\mathbf{r}_{b,t}$ is not identical to a single global operator on the full residual vector $\mathbf{r}_t$. Figure 2 contrasts the global and mixture pipelines.

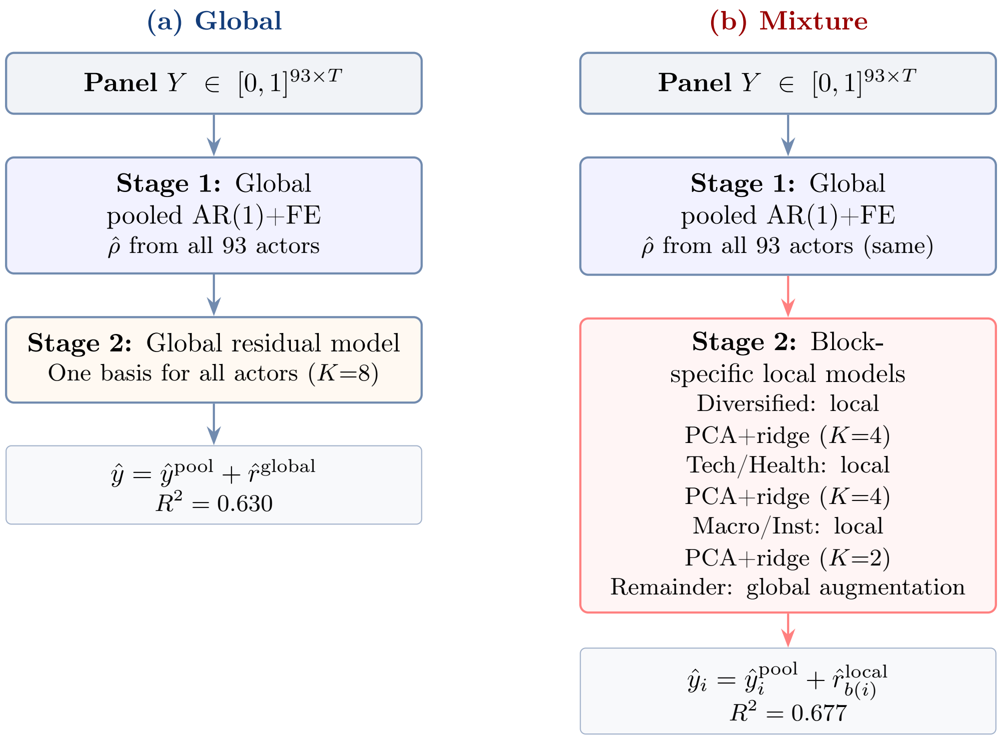

Figure 2: Architecture comparison. (a) The global pipeline applies one residual model to all 93 actors. (b) The mixture pipeline uses the same global Stage 1 but routes actors to block-specific local models in Stage 2, with economically pre-specified blocks. Stage 1 is identical; only the Stage 2 routing differs.

4 When Global Models Misalign

4.1 Full-Panel Result

Table 5 compares five architectures on the full 93-actor panel; Figure 3 shows the per-window comparison. The mixture architectures (M1, M2) beat the global always-on augmentation (G1) in every window.

Table 5: Five architectures on the full 93-actor panel (10 rolling OOS windows, 2015–2024). S1 applies pooled-only (no augmentation) to local blocks and global augmentation to the remainder. M1 and M2 apply local Ridge or local PCA+ridge to local blocks. W denotes the number of windows (out of 10) with a positive gain over G1.

#	Architecture	R^2	Δ vs G1	t	p	CI	W
G0	Pooled-only	0.591	−0.039	−7.14	<0.001	$[-0.048, -0.029]$	0/10
BA	Block-specific $\hat{\rho}_b$+FE	0.611	−0.019	−3.72	0.005	$[-0.028, -0.009]$	0/10
G1	Global always-on	0.630	—	—	—	—	—
S1	Selective-off	0.599	−0.031	−8.55	<0.001	$[-0.037, -0.024]$	0/10
M1	**Mixture (Ridge)**	**0.669**	**+0.039**	**7.80**	**<0.001**	$[\mathbf{+0.030}, \mathbf{+0.049}]$	**10/10**
M2	**Mixture (PCA+ridge)**	**0.677**	**+0.047**	**7.76**	**<0.001**	$[\mathbf{+0.036}, \mathbf{+0.058}]$	**10/10**

The block-specific $\hat{\rho}_b$+FE baseline (BA) — which allows each block its own persistence parameter but has no Stage 2 — achieves $R^2 = 0.611$, well below M2 at 0.677 ($\Delta = +0.066$, $t = 7.92$, CI $[+0.052, +0.082]$, 10/10 windows). The gain is not from heterogeneous persistence alone: the local Stage 2 residual dynamics add genuine value beyond what block-specific $\hat{\rho}_b$ captures.

The selective-off architecture (S1) is *worse* than global ($\Delta = -0.031$), demonstrating that the mixture gain is not from removing suboptimal augmentation. The local models genuinely add predictive value: M1 exceeds S1 by +0.070 ($t = 16.12$, $p < 0.001$).

The primary comparison (M2 vs G1) is robust to dependence correction. With Newey–West HAC variance (bandwidth 1–3), the DM-HAC statistic ranges from 6.84 to 7.38 ($p < 0.001$ in all cases). A moving-block bootstrap (block lengths 2 and 3) yields CIs of $[+0.035, +0.059]$ and $[+0.036, +0.059]$, both excluding zero. The conclusion is invariant to the treatment of window-level dependence.

Whether a direct forecast combination (Bates and Granger, 1969) of global-basis models could replicate the mixture gain is unlikely but not formally excluded. M2 exceeds the best component (G1) in every window by +0.025 to +0.074, and the mixture estimates block-specific factor structure ($\hat{U}_b$, $\hat{A}_b$) from within-block residuals not contained in any weighted average of global-basis forecasts—the information sets differ structurally, not just in weighting. A formal out-of-sample forecast-combination baseline with training-estimated weights remains a natural further comparison.

4.2 Per-Block Decomposition

Table 6 and Figure 4 reveal the mechanism. The global basis causes cross-block interference that manifests differently across blocks.

Table 6: Per-block R^2 by architecture. The diversified sector is degraded by global augmentation ($0.392 < 0.415$ pooled-only). The tech/health block gains most from local treatment. Note: per-block R^2 uses block-specific total sum of squares as denominator, so the N-weighted average of per-block R^2's does not equal the full-panel R^2 in Table 5.

Block	N	G0 (pooled)	G1 (global)	S1 (sel-off)	M1 (Ridge)	M2 (PCA+r)
Diversified	23	0.415	0.392	0.415	0.461	0.449
Macro/Institutional	11	0.600	0.649	0.600	0.692	0.689
Tech/Health	25	0.554	0.681	0.554	0.764	0.808
Remainder	34	0.622	0.646	0.646	0.646	0.646

Three patterns emerge. First, the diversified sector is the clearest case of cross-block interference:

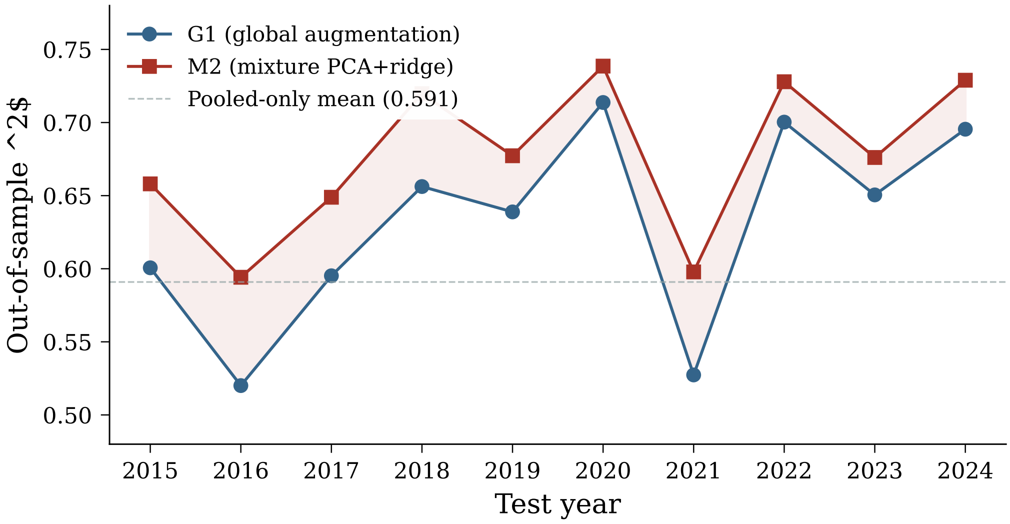

Figure 3: Out-of-sample R^2 by test year for global always-on augmentation (G1) and the mixture architecture (M2). M2 exceeds G1 in all 10 windows. Horizontal reference lines show pooled-only and per-actor AR(1) baselines.

global augmentation *reduces* R^2 from 0.415 (pooled-only) to 0.392. The global basis, dominated by technology/healthcare and macro modes, injects noise into predictions for this heterogeneous sector. Local Ridge recovers $R^2 = 0.461$.

Second, the tech/health block shows the largest absolute gain. Global augmentation already helps substantially (+0.127 over pooled), but local PCA+ridge with $K = 4$ nearly doubles the gain to $R^2 = 0.808$. The within-block co-movement structure is captured far more efficiently by a local 4-component model than by a global 8-component model that mixes in macro and diversified dynamics.

Third, the remainder block (energy, industrials, financials) is unchanged by design—the same global augmentation is applied in all architectures. The global basis serves these sectors well; local decomposition is not needed.

4.3 Geometric Description

At the rank K=4 used for block-level comparison, the global spectral basis rotates at 31.5° per quarter (geodesic distance on the Grassmannian); within-block rotation ranges from 14° to 28° (per-block values available in the replication archive). However, a matched-size random sub-panel control (200 draws per block size, same K and metric) shows the reduction is *not statistically significant*: Tech/Health $p = 0.080$; Macro/Institutional $p = 0.145$; Diversified $p = 0.305$. The lower within-block rotation is largely a mechanical effect of smaller panel size producing more stable estimated bases. The geodesic evidence is therefore descriptive context, not a demonstrated mechanism—the forecasting result rests on the actual R^2 comparison (Table 5; placebo $z = 7.56$, Table 7), not on the geometric story.

5 Validation

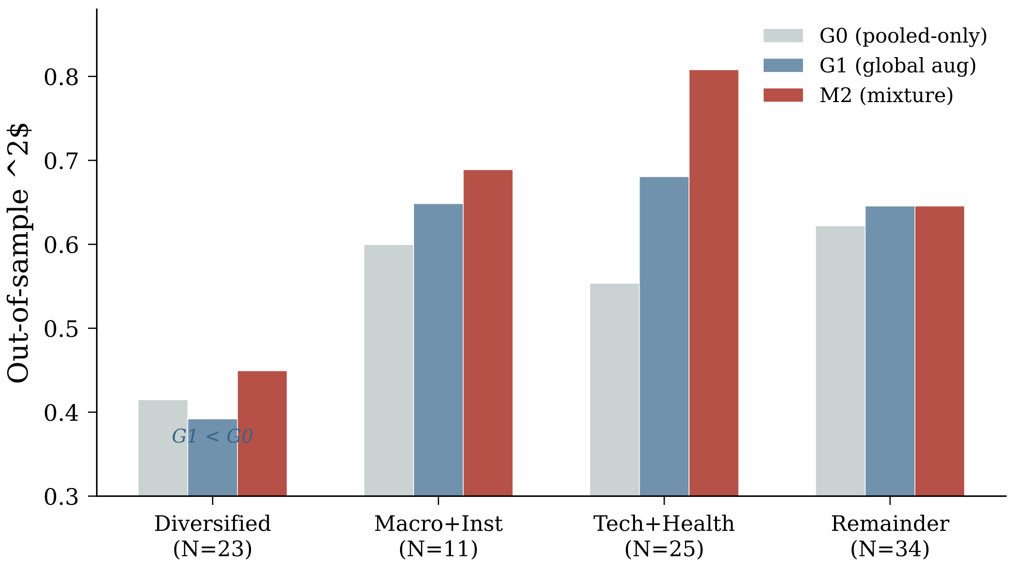

Figure 4: Per-block R^2 under three architectures: pooled-only (G0), global augmentation (G1), and mixture PCA+ridge (M2). The diversified sector is degraded by global augmentation (G1 < G0); the tech/health block benefits most from local treatment.

5.1 Placebo Test

The strongest potential objection to block-specific modelling is overfitting the block structure. We address this with a placebo benchmark (Table 7, Figure 5): 50 random partitions of the 93 actors into blocks of exactly the same sizes as the real partition (23, 11, 25, 34). For each random partition, the full mixture pipeline (M2) is run with identical local model specifications.

Table 7: Placebo test: 50 random partitions vs real economic blocks.

Statistic	Value
Real economic blocks Δ	+0.047
Placebo mean Δ	−0.006
Placebo std	0.007
Placebo maximum	+0.012
Placebo 99th percentile	+0.011
z-score (real vs placebo)	7.56
Placebo p-value	<0.020 (0/50; Monte Carlo bound 1/51)

Random blocks degrade performance on average ($\Delta = -0.006$) because local estimation noise dominates when blocks are arbitrary. No random partition approaches the real gain. Each placebo uses the same 10 evaluation windows, same block sizes, and same local model specification—the only difference is actor assignment. The placebo benchmark controls for small-sample bias by design.

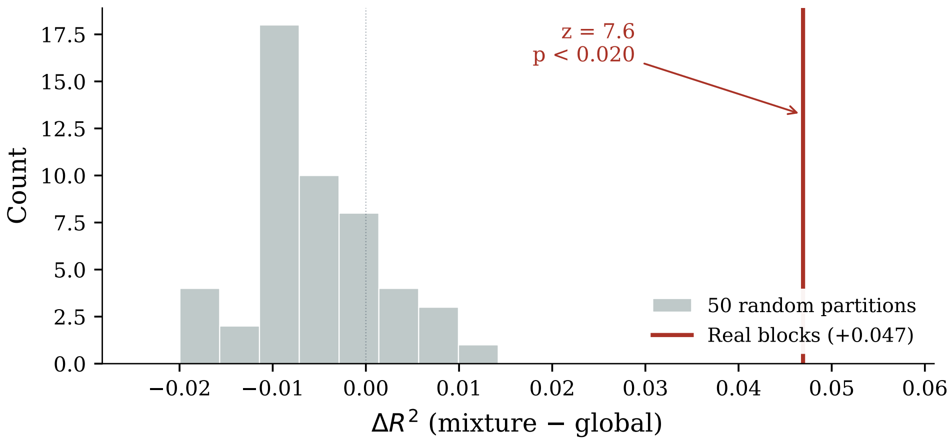

Figure 5: Distribution of ΔR^2 (mixture − global) across 50 random block partitions of identical sizes (grey histogram) versus the real economic partition (vertical red line at +0.047). Random blocks produce a mean gain of −0.006; no random partition exceeds +0.012. The real gain exceeds the placebo distribution by $z = 7.56$ (Monte Carlo $p < 0.020$).

5.2 Cross-Panel Validation

Table 8: Cross-panel validation. The mixture gain requires data-type heterogeneity.

Panel	N	Heterogeneous?	Global R^2	Mixture R^2	Δ
93-actor multilayer	**93**	**Yes**	**0.630**	**0.677**	**+0.047**
146-firm CapEx/Revenue	146	No	0.745	0.743	−0.002
270-actor multi-ratio	270	Moderate	0.753	0.754	+0.001

The mixture architecture does not improve prediction on homogeneous firm-only panels. The 146-firm panel (all firms, one ratio, one normalisation) shows no benefit: the global basis serves all actors equally well when the panel is homogeneous.[1] The 270-actor panel (all firms, two ratio types) also shows no benefit—the ratio-type distinction is not severe enough to warrant local decomposition. The gain requires *data-type heterogeneity*: mixing fundamentally different time series (trending macro indices and mean-reverting firm ranks) in a single panel.

5.3 Leave-One-Window-Out Block Selection

A potential concern is that the choice of which blocks receive local treatment was informed by the same evaluation windows used to measure the gain. To address this, a strict leave-one-window-out (LOWO) procedure is applied: for each test year w, the block selection uses per-block R^2 from the *other* 9 windows only. A block receives local treatment if $\bar{\Delta}_{-w} = \overline{R^2_{\text{local}} - R^2_{\text{global}}} > 0$ across the held-out windows.

The result is unambiguous: all 10 windows select the same three blocks (Diversified, Macro/Institutional, Tech/Health). Since LOWO selects the identical partition every window, the resulting forecasts are the same as the fixed-partition mixture by construction—the gain is +0.047 with the same

[1] A separate spectral architecture—standalone DMD+Kalman with K=2 operating directly on the EWM-demeaned panel without a pooled Stage 1—provides a small gain over per-actor AR(1) on the 146-firm panel (+0.028 R^2, 9/10 windows positive). This operates through a different mechanism (low-rank persistence compression rather than cross-block interference removal) and is the subject of a separate methodological investigation.

CI and significance. The per-block gain signal is so consistent across windows that no window's inclusion or exclusion changes the block selection.

5.4 Robustness: Dropping the Diversified Sector

The diversified sector is the most heterogeneous block and the one where global augmentation most clearly degrades prediction. To test whether the headline gain depends on this specific block, the entire pipeline is re-estimated on the 70-actor panel after removing all diversified-sector actors. On this reduced panel (blocks: Macro/Institutional N=11, Tech/Health N=25, Remainder N=34), the mixture gain is $\Delta = +0.035$ ($t = 10.37$, CI $[+0.029, +0.042]$, 10/10 windows positive)—75% of the original gain. The headline result does not depend primarily on the diversified sector: 75% of the gain survives its removal, sustained primarily by the tech/health block. The remaining 25% reflects the diversified block's specific contribution.

5.5 Economic Content

Gap-revision regressions on the global augmentation model confirm that forecast gaps predict subsequent investment revisions with actor fixed effects, consistent with the forecasts capturing economically meaningful variation rather than statistical noise. Augmented gaps are less predictable than pooled gaps, consistent with signal absorption. A full economic validation of the mixture architecture—including portfolio sorts, latency-adjusted backtests, and economic magnitude assessment—is beyond the scope of this methodological paper and is an important direction for future applied work.

5.6 Train-Only Causality Audit

Block assignments use static economic metadata (sector and layer labels from the actor registry). Local models are re-estimated each quarter from training data only: PCA basis from training residuals, Ridge VAR from training factors. The global Stage 1 and Stage 2 (for the remainder block) are re-estimated quarterly as each test quarter's data becomes available. No test-quarter information enters any model component.

6 What Does Not Work and Why

The heterogeneity finding emerged from a systematic programme that tested—and ruled out—four alternative explanations for the $R^2 \approx 0.630$ ceiling. Each negative result narrows the explanation space, leaving block-specific decomposition as the only surviving path to improvement. Table 9 summarises the programme.

Table 9: Summary of alternative explanations tested and ruled out.

Alternative	Test	Result	Implication
Better spectral method	9 models, 3 complexity classes	DMD $\approx$ PCA $\approx$ Ridge ($\rho > 0.98$)	Architecture, not method
Forecastable global geometry	10 geometric models on Grassmannian	All worse than persistence	Rotation is estimation noise
Target formulation	7 target variants	Max $\lvert\Delta\text{Gain}\rvert = 0.018$	Ceiling is target-invariant
Conditional gating	5 gating policies	All worse than always-on	Augmentation is unconditional

6.1 The Method Does Not Matter

Section 3.4 established method equivalence globally. The same result holds locally: within each of the 9 blocks tested, local DMD never outperforms local PCA+ridge (mean $\Delta = -0.028$, worst -0.059). The method is irrelevant at every granularity.

6.2 Global Geometry Is Not Forecastable

The global spectral basis rotates at $49.2° \pm 16.8°$ per quarter at the model's operating rank $K{=}8$, dominated by a single principal angle ($\theta_1 = 45.8°$). Despite the structural regularity, the rotation is temporally unpredictable: magnitude ACF(1) $= -0.07$, Ljung-Box $p = 0.29$, direction cosine $= 0.047$. All 10 geometric prediction models (Grassmannian extrapolation, projector averaging, Karcher mean, HS-linear regression) perform strictly worse than persistence. The rotation is consistent with finite-sample estimation noise. Full diagnostics appear in Appendix F; structural spectral analysis (mode loadings, variance decomposition) is in Appendix C.

6.3 The Target Is Not the Problem

Seven target formulations (percentile ranks, normal quantiles, winsorised z-scores, sector-relative ranks, moving averages, first differences) produce augmentation gains within ± 0.018 of the baseline (excluding first differences, which show a larger gain of $+0.051$ but with absolute R^2 of only 0.047). Split-half reliability yields a noise-corrected ceiling of ~ 0.765, suggesting some headroom from measurement noise, but no target reformulation closes the gap. Appendix D provides the full comparison.

6.4 Conditional Gating Does Not Help

Five gating policies (NCD-based, dispersion-based, persistence-based, effective-rank-based, and combined) all produce *lower* R^2 than always-on augmentation. The best gate (dispersion) loses 0.014 versus always-on. Augmentation is unconditionally beneficial—no train-only diagnostic identifies quarters where the second stage should be deactivated. Appendix E provides the full comparison.

7 Discussion

7.1 Pool Globally, Decompose Locally

The paper's architectural principle is summarised in one sentence: *estimate persistence globally (where pooling helps), estimate residual dynamics locally (where pooling introduces interference).* Whether this principle applies beyond the specific panel studied here is an empirical question: the cross-panel validation (Table 8) shows null effects on two homogeneous firm-only panels, confirming the gain requires data-type heterogeneity. Panels mixing fundamentally different actor types in a common factor structure are natural candidates for further testing. The geodesic decomposition (global rotation $\gg$ local rotation) provides a suggestive diagnostic for when local decomposition may be warranted, though its predictive value for new panels has not been validated.

7.2 Connection to Heterogeneous Factor Model Literature

The finding connects to several strands of the econometric literature on panel heterogeneity. Bonhomme and Manresa (2015) study grouped patterns of heterogeneity in panel models, estimating group-specific intercepts and slopes. Su et al. (2016) develop methods for identifying latent structures in panel data. Ando and Bai (2017) cluster financial time series using a panel data approach. Ke et al. (2015) introduce homogeneity pursuit—testing which coefficients can be pooled and which must be group-specific. Our approach differs in using pre-specified economic blocks rather than data-driven grouping, but the geodesic decomposition could inform automated block discovery in future work.

7.3 Limitations

This paper is a purely empirical forecasting study. The evaluation operates in a finite-sample regime ($N = 93$, $T \approx 20$ per window, $K = 2$–8 modes) where classical large-N or large-T asymptotics do not directly apply. The validity of the comparisons rests on the rolling out-of-sample protocol, not on asymptotic guarantees. Whether the block-specific mixture estimator (Equation 2) enjoys formal consistency properties under a well-defined asymptotic regime is an open question.

A block-specific pooled AR(1)+FE with block-specific $\hat{\rho}_b$ (no Stage 2) achieves $R^2 = 0.611$—substantially below the mixture at 0.677 (Table 5). This confirms the gain is from heterogeneous *residual* structure, not just heterogeneous persistence. A single-stage Ridge with block-dummy interactions or a forecast-combination scheme with block-level weights (Bates and Granger, 1969) remain natural additional comparisons for future work.

Several further limitations qualify the findings:

1. **Small evaluation sample.** Ten out-of-sample windows is standard in rolling-window evaluation but provides limited statistical power. The placebo test mitigates this concern by using the same windows.
2. **Block selection informed by exploratory analysis.** The block identities (diversified, macro/institutional, tech/health) are economic classifications, but the *selection* of which blocks receive local treatment was guided by preliminary within-sample diagnostics. Specifically, 10 candidate blocks were evaluated (6 raw sector blocks, 2 merged sector blocks, 2 layer blocks), of which 3 were selected based on per-block local-vs-global R^2 comparisons. The LOWO test controls for selection conditional on the candidate set; the placebo controls for the block sizes. Neither fully controls for the search over the 10 candidates.
3. **Scope limited to heterogeneous panels.** The gain does not replicate on homogeneous firm-only panels. The finding is a precise scope condition, not a universal architectural improvement.
4. **Quarterly frequency only.** Higher-frequency data may reveal different dynamics.
5. **Block boundary sensitivity untested.** Robustness to marginal reclassification of borderline actors between blocks has not been tested.
6. **Modest absolute improvement.** The +0.047 R^2 gain, while statistically robust ($z = 7.56$), is modest in absolute terms.
7. **FRED min-max look-ahead.** The 7 macro actors use full-sample min-max normalisation, introducing a mild look-ahead in target values. This does not affect the 82 firm actors (cross-sectional ranks are contemporaneous). The main result is driven by the tech/health block (no FRED actors), so the finding is not dependent on the macro normalisation.
8. **Data-timing latency.** Quarterly firm ratios from SEC filings are typically available 30–45 days after quarter-end, while macro indicators are available immediately. The evaluation assumes all targets are available at the quarterly boundary. A real-time implementation

would need to account for filing lags.

7.4 Configuration for Short Training Windows

The main results use a 5-year training window, but practitioners often face shorter histories. Table 10 reports the mixture gain across (T, K_b) pairs on the 93-actor panel.

Table 10: Mixture gain $\Delta(M2 - G1)$ by training-window length T and local rank K_b on the 93-actor panel (10 rolling OOS windows each). The paper's operating point (T=5yr, K_b=4) is marked with $\star$.

T (yr)	K_b=2	K_b=3	K_b=4	K_b=6
2	+0.083	+0.090	**+0.097**	+0.095
3	+0.035	+0.049	+0.053	**+0.054**
5	+0.029	+0.045	+0.047*	+0.043

Two findings emerge. First, K_b=4 is near-optimal at every tested training-window length. Lower values (K_b=2) lose 0.014–0.018 R^2 versus K_b=4 at all T; K_b=6 is worse on aggregate because the smaller Macro/Institutional block (N_b=11) cannot support six modes, even though the tech/health block alone benefits from K_b=6. The pre-specified K_b values in Table 4 are robust to the choice of T.

Second, the mixture gain *increases* as the training window shrinks: +0.047 at T=5yr, +0.054 at T=3yr, +0.097 at T=2yr—roughly doubling. The mechanism is sample-size-dependent: at short T, the 93-actor global basis is poorly estimated from few observations, while the local 11–25-actor bases retain comparable conditioning because they target lower-dimensional subspaces. Cross-block interference in the global model worsens as T shrinks, so the architectural gain from removing it grows correspondingly. The ridge penalty on the local VAR coefficients prevents the underdetermined basis estimates (N_b=25, K_b=4, T=8 quarters) from translating into unstable forecasts.

This finding contradicts a pre-registered prediction that lower K_b would be needed at shorter T; the opposite is true.[2] For practitioners with limited training history, the heterogeneity-aware mixture architecture is more valuable, not less.

7.5 Extensions

Three extensions follow naturally. First, automated block discovery via cross-validated clustering on training-window loadings, replacing the pre-specified partition with a data-driven one. Second, higher-frequency data (monthly or daily), where the basis rotation may become temporally predictable and geometric methods may add value. Third, change-based prediction targets: preliminary evidence shows augmentation gain on first-differences is 2.5× larger than on levels, suggesting the spectral dynamics align better with change structure.

8 Conclusion

This paper studies cross-sectional investment forecasting in heterogeneous panels containing actors with fundamentally different dynamics. Three findings emerge.

[2] A post-hoc $T \times K_b$ grid was conducted with five pre-registered predictions, all of which failed in directions that strengthen the architectural finding. Full pre-registration and results in the replication archive.

First, a two-stage architecture—global pooled persistence followed by regularised residual dynamics—improves predictive R^2 by 1.7–3.6 percentage points over standard baselines across three panels and invariant to the choice of second-stage method (PCA, DMD, or Ridge).

Second, global second-stage pooling can degrade prediction for actor subgroups whose dynamics are misrepresented by the shared basis. Descriptively, within-block basis rotation is lower than global rotation, though a matched-size random sub-panel control shows this reduction is not statistically distinguishable from a sample-size effect. Regardless of the geometric mechanism, the forecasting result is clear: a heterogeneity-aware mixture architecture with block-specific local residual models improves full-panel R^2 by +0.047 over the best global augmentation, validated by a placebo benchmark ($z = 7.56$), dependence-robust inference (DM-HAC $t > 6.8$), and consistency across all 10 evaluation windows.

Third, the improvement requires data-type heterogeneity—mixing macro indices with firm-level cross-sectional ranks—and does not arise on homogeneous panels. This scope condition, together with the geodesic diagnostic and method-equivalence finding, provides an empirical characterisation of when local decomposition adds forecasting value on this class of panels.

The design principle is simple: *pool globally for what is shared, decompose locally for what is heterogeneous.*

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A DMD Mathematics

Dynamic Mode Decomposition (DMD) provides a rank-K approximation of the linear propagator mapping consecutive cross-sectional snapshots. Given a snapshot matrix $X = [x_1, \dots, x_{T-1}]$ and $Y = [x_2, \dots, x_T]$ with $x_t \in \mathbb{R}^N$, the truncated SVD $X = U_r \Sigma_r V_r^*$ yields the reduced propagator $\tilde{A} = U_r^* Y V_r \Sigma_r^{-1} \in \mathbb{R}^{K \times K}$, where $K \leq \min(N, T-1)$ is the truncation rank. The eigendecomposition $\tilde{A}W = W\Lambda$ produces DMD eigenvalues λ_k encoding per-mode growth rate ($|\lambda_k|$) and oscillation frequency ($\arg \lambda_k$). For forecasting, the relevant objects are the left singular vectors U_r (the orthonormal basis spanning the dominant K-dimensional subspace) and the diagonal or full matrix $\tilde{A}$ (the transition dynamics). In the paper's two-stage architecture, these are applied to Stage 1 residuals, not to the raw panel. Spectral radius clipping $\tilde{A} \leftarrow \tilde{A} \cdot \min(1, 0.99/\max_k |\lambda_k(\tilde{A})|)$ ensures stability.

DMD-Based Kalman Filter

The DMD basis U_r and transition $\tilde{A}$ define a linear state-space model in K-dimensional modal coordinates. Let $\boldsymbol{\alpha}_t \in \mathbb{R}^K$ denote the modal state at time t. The filter operates as follows.

Initialisation. $\boldsymbol{\alpha}_0 = \mathbf{0}$, $P_0 = I_K$, $Q_0 = q_0 \cdot I_K$ (with $q_0 = 0.5$).

Prediction step.

$$\boldsymbol{\alpha}_{t|t-1} = F\,\boldsymbol{\alpha}_{t-1|t-1}, \qquad P_{t|t-1} = F\,P_{t-1|t-1}\,F^\top + Q_{t-1} \tag{3}$$

where $F = \mathrm{clip}_{\mathrm{SR}}(\tilde{A}, 0.99)$ is the spectral-radius-clipped transition (diagonal $\mathrm{diag}(\tilde{A})$ for the default model; full $\tilde{A}$ for max-performance).

Observation model. The predicted residual vector is $\hat{r}_t = U_r\,\boldsymbol{\alpha}_{t|t-1} + \bar{r}$, where $\bar{r}$ is the EWM mean of training residuals (half-life 12 quarters).

Measurement noise (spherical regularisation). The observation noise covariance is set to

$$R = \sigma_\perp^2\, I_N, \qquad \sigma_\perp^2 = \frac{1}{NT}\sum_{t,i}\big(r_{i,t} - [U_r U_r^\top r_t]_i\big)^2 \tag{4}$$

i.e., the mean squared projection residual of the training data onto the K-dimensional subspace. This spherical form assumes homogeneous noise across actors and avoids estimating an $N \times N$ covariance with $T \ll N$. It is the key regularisation that makes the Kalman filter well-conditioned in the $N > T$ regime.

Update step.

$$S_t = U_r\,P_{t|t-1}\,U_r^\top + R \tag{5}$$
$$K_t = P_{t|t-1}\,U_r^\top\,S_t^{-1} \tag{6}$$
$$\boldsymbol{\alpha}_{t|t} = \boldsymbol{\alpha}_{t|t-1} + K_t\big(\tilde{r}_t - U_r\,\boldsymbol{\alpha}_{t|t-1}\big) \tag{7}$$
$$P_{t|t} = (I_K - K_t\,U_r)\,P_{t|t-1} \tag{8}$$

where $\tilde{r}_t = r_t - \bar{r}$ is the demeaned observed residual.

Adaptive process noise. Q is updated each quarter via exponential smoothing of the innovation outer product:

$$Q_t = (1-\lambda_Q)\,Q_{t-1} + \lambda_Q\,\nu_t\nu_t^\top, \qquad \nu_t = \boldsymbol{\alpha}_{t|t} - \boldsymbol{\alpha}_{t|t-1} \tag{9}$$

with $\lambda_Q = 0.3$, symmetrised and regularised: $Q_t \leftarrow \frac{1}{2}(Q_t + Q_t^\top) + 10^{-6} I_K$.

Rolling basis update. At each quarterly step, the DMD basis U_r and transition $\tilde{A}$ are re-estimated from the expanding training data. When the basis changes, the Kalman state is re-projected ($\boldsymbol{\alpha} \leftarrow U_r^{\text{new}\top} \tilde{r}_t$), and P, Q are reset to their initial values.

Key design choices. (i) The spherical R (Eq. 4) makes the filter well-conditioned for $N = 93$, $K = 8$, $T \approx 20$, where estimating a full $N \times N$ observation covariance is infeasible. (ii) The adaptive Q (Eq. 9) allows the process noise to track regime changes without manual tuning. (iii) Spectral radius clipping at 0.99 prevents explosive modal dynamics. (iv) The quarterly basis re-estimation tracks evolving cross-sectional structure but discards accumulated Kalman state, resetting each quarter.

B Standalone Transition Diagnostic

The standalone spectral model (DMD-based Kalman filter with $K = 8$ modes) achieves $R^2 = 0.415$ with the default near-identity transition $F = 0.99I$ versus $R^2 = 0.594$ for per-actor AR(1). Replacing F with the diagonal of $\tilde{A}$ (per-mode eigenvalue dynamics) improves to $R^2 = 0.483$; the full reduced propagator $\tilde{A}$ yields $R^2 = 0.486$. The gap versus AR(1) is fundamental: a single spectral basis cannot replicate actor-specific persistence. The two-stage architecture resolves this by separating persistence (Stage 1) from cross-sectional structure (Stage 2).

C Structural Spectral Analysis

The 93-actor panel's spectral structure includes 8 stable DMD modes with eigenvalue magnitudes $|\lambda_k| \in [0.87, 0.97]$ and substantial oscillatory components ($\theta \approx 44$–$91°$). Modal $R^2 = 0.69$ (in-sample) confirms the structure is real. Mode interpretation from training-window loadings shows that modes 1–2 load on technology/healthcare versus financials (sector rotation), while macro loadings are negligible after removing shared persistence—consistent with the heterogeneity finding.

D Target Sensitivity Details

Seven target formulations tested: percentile ranks (baseline), normal quantiles ($R^2 = 0.610$, gain +0.009), winsorised z-scores ($R^2 = 0.628$, gain +0.018), sector-relative ranks ($R^2 = 0.595$, gain +0.016), 2Q moving average ($R^2 = 0.811$, gain +0.016), 4Q moving average ($R^2 = 0.934$, gain +0.014), first differences ($R^2 = 0.047$, gain +0.051). The first-differences result shows augmentation gain is 2.5× larger for changes than levels, but absolute R^2 is low. Split-half reliability: $\rho = 0.513$, Spearman-Brown reliability = 0.678, noise-corrected ceiling ≈ 0.765.

E Gating Policy Details

Five quarter-level causal gating policies compared to always-on augmentation: NCD gate ($\Delta = -0.015$), dispersion gate ($\Delta = -0.014$), persistence gate ($\Delta = -0.019$), effective-rank gate ($\Delta = -0.033$), combined gate ($\Delta = -0.028$). All worse than always-on. The augmentation is unconditionally beneficial.

F Rotation Diagnostics and Predictability

The global spectral basis rotates at $49.2° \pm 16.8°$ per quarter (geodesic distance at the model's operating rank $K = 8$). At $K = 4$ (used for the within-block comparison in Section 4.3, where block sizes preclude $K = 8$), the global geodesic is 31.5°. The rotation is dominated by $\theta_1 = 45.8°$ (93% of geodesic distance). Minor angles θ_4–θ_8 ($< 2°$) show significant autocorrelation (ACF > 0.5) but contribute negligibly to the total rotation—a persistence-relevance trade-off. Six representative subspace prediction models tested: Grassmannian extrapolation ($\Delta = +0.021$ vs persistence, *worse*), tangent-space AR (+0.077, worse), angle AR (+0.080, worse), constant velocity (+0.149, worse), Euclidean projector average (+0.371, worse), Karcher mean (+0.383, worse). No model beats persistence; the rotation is temporally unpredictable.